Trade Policy Uncertainty and the Chilean Peso

A two-period portfolio equilibrium model

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A conditional risk-compensation mechanism
and an empirical plan

Research question and conditional answer

Can higher trade policy uncertainty weaken the peso even when its expected terminal dollar value is unchanged?

- Risk-averse investors choose between dollar and peso assets.
- The current exchange rate adjusts to clear a fixed supply of peso assets.
- The direction depends on the currency risk investors collectively bear.

Main result

A mean-preserving increase in terminal peso risk lowers its current dollar price when the pricing investor sector has **positive net peso exposure**.

A depreciation alone cannot distinguish this effect from adverse news about the expected future value of the peso.

Related approach: Gabaix and Maggiori (2015), *QJE*. The construction here is a separate two-state partial-equilibrium model.

Timing, exchange-rate quotation, and assets

Date 0: information and trading

- Investors observe TPU-related information u.
- They trade at the market-clearing exchange rate s_0.

Date 1: uncertainty resolves

- Assets pay in their own currency.
- Terminal wealth is evaluated in dollars.

$$s_0 = \text{USD per CLP}, \qquad E_0 = \frac{1}{s_0} = \text{CLP per USD}.$$

Asset	Date-0 dollar cost	Date-1 dollar payoff
Dollar asset	1	R^*
One unit of peso asset	s_0	$R\widetilde{s}_1$

$R, R^* > 0$ are fixed gross interest factors. Peso depreciation means s_0 falls and E_0 rises.

Terminal currency risk and the uncertainty experiment

Investors share the conditional two-state distribution

$$\widetilde{s}_1 = \mu(u) + \sigma(u)\varepsilon, \qquad \Pr(\varepsilon = +1 \mid u) = \Pr(\varepsilon = -1 \mid u) = \tfrac{1}{2}.$$

$$0 < \sigma(u) < \mu(u), \qquad \mathbb{E}_0[\widetilde{s}_1] = \mu(u), \qquad \mathrm{Var}_0(\widetilde{s}_1) = \sigma(u)^2.$$

Pure uncertainty experiment

Hold $\mu(u) = \mu_0$ fixed and let $\sigma'(u) > 0$. The two terminal values move farther apart around the same mean.

- The restriction $\mu > \sigma$ keeps both terminal exchange rates positive.
- A news-based TPU index need not represent a pure dispersion shock.
- Holding $\mathbb{E}_0[\widetilde{s}_1]$ fixed does not hold $\mathbb{E}_0[1/\widetilde{s}_1]$ fixed.

TPU measurement: Caldara et al. (2020), *Journal of Monetary Economics*.

Investors, budgets, and background exposure

Investor i has CARA utility $U_i(W) = -\exp(-\gamma_i W)$, with $\gamma_i > 0$. Borrowing, lending, and short positions are unrestricted.

$$b_i + s_0 n_i = w_i, \qquad \widetilde{W}_{i1} = R^* b_i + R(n_i + h_i)\widetilde{s}_1 + y_i.$$

- w_i: initial dollar wealth; b_i: dollars invested in the safe asset.
- n_i: peso-asset units; y_i: safe terminal dollar income.
- Rh_i: signed terminal peso cash flow. Income has $h_i > 0$, liabilities have $h_i < 0$.

$$H = \sum_i h_i, \qquad T = \sum_i \frac{1}{\gamma_i} > 0, \qquad K = B + H.$$

$B > 0$ is net asset supply to the sector; T is total risk tolerance. After market clearing, K is the sector's total net peso exposure.

Portfolio choice has an exact certainty equivalent

Define the expected excess dollar payoff and payoff dispersion:

$$a = R\mu - R^* s_0, \qquad d = R\sigma > 0.$$

Substitution into wealth gives $\widetilde{W}_{i1} = C_i + n_i a + d(n_i + h_i)\varepsilon$, where $C_i = R^* w_i + y_i + R h_i \mu$.

$$CE_i(n_i) = C_i + n_i a - \frac{1}{\gamma_i} \ln \cosh\big[\gamma_i d(n_i + h_i)\big].$$

$$CE_i'(n_i) = a - d \tanh\big[\gamma_i d(n_i + h_i)\big],$$
$$CE_i''(n_i) = -\gamma_i d^2 \operatorname{sech}^2\big[\gamma_i d(n_i + h_i)\big] < 0.$$

The first-order condition is sufficient for a unique finite optimum whenever the price satisfies the strict no-arbitrage bounds.

Asset demand and admissible prices

A finite optimum requires $|a| < d$, or

$$\frac{R(\mu - \sigma)}{R^*} < s_0 < \frac{R(\mu + \sigma)}{R^*}.$$

Inverting the first-order condition yields

$$n_i^d(s_0, u) = \frac{1}{\gamma_i d} \operatorname{artanh}\left(\frac{a}{d}\right) - h_i.$$

- A lower current price raises the expected excess payoff and asset demand.
- Background peso income reduces asset demand one for one.
- Peso liabilities raise hedging demand.

At either bound, unrestricted positions prevent a finite optimum.

$$\operatorname{artanh}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1.$$

Market clearing determines a unique current exchange rate

Peso-asset clearing requires $\sum_i n_i = B$. Summing demands gives

$$\frac{T}{d}\,\operatorname{artanh}\left(\frac{a}{d}\right) - H = B \quad \Longrightarrow \quad a = d\tanh\left(\frac{Kd}{T}\right).$$

Let $z = KR\sigma/T$. The exact equilibrium is

$$s_0 = \frac{R}{R^*}\left[\mu - \sigma\tanh\left(\frac{KR\sigma}{T}\right)\right].$$

- **Existence:** $|\tanh(z)| < 1$ places this positive price strictly inside the admissible interval.
- **Uniqueness:** aggregate demand decreases continuously from $+\infty$ to $-\infty$ across that interval.

This closes the asset market. Terminal fundamentals, goods markets, and monetary policy remain outside the model.

Risk compensation and equilibrium risk sharing

The expected excess dollar payoff per unit of the peso asset is

$$\rho \equiv R\mu - R^* s_0 = R\sigma \tanh(z).$$

The expected excess gross dollar return is

$$\mathbb{E}_0\left[\frac{R\widetilde{s}_1}{s_0}\right] - R^* = \frac{\rho}{s_0}.$$

Equilibrium allocations satisfy

$$n_i^{\text{eq}} = \frac{K}{\gamma_i T} - h_i, \qquad n_i^{\text{eq}} + h_i = \frac{K}{\gamma_i T}.$$

- Net currency risk is shared in proportion to absolute risk tolerance.
- $K > 0$ requires positive compensation. With $K < 0$, the asset hedges aggregate peso obligations and compensation is negative.

Pure uncertainty: the sign depends on net exposure

Fix μ, R, R^*, K, T and let $\sigma'(u) > 0$. Define

$$G(z) = \tanh(z) + z\,\mathrm{sech}^2(z), \qquad z = \frac{KR\sigma}{T}.$$

$$\frac{\partial s_0}{\partial u} = -\frac{R}{R^*}\sigma'(u)G(z), \qquad \frac{\partial \rho}{\partial u} = R\sigma'(u)G(z).$$

Both terms in $G(z)$ have the sign of K.

Net exposure	s_0 (USD/CLP)	E_0 (CLP/USD)	Peso response
$K > 0$	Falls	Rises	Depreciation
$K = 0$	Unchanged	Unchanged	No price effect
$K < 0$	Rises	Falls	Appreciation

With $K > 0$, the lower purchase price compensates investors for greater risk.

Expected-value news can reinforce or offset the risk effect

Allow μ to change with u, while keeping rates, exposure, and risk tolerance fixed:

$$\frac{\partial s_0}{\partial u} = \frac{R}{R^*}\left[\underbrace{\mu'(u)}_{\text{expected terminal value}} - \underbrace{\sigma'(u)G(z)}_{\text{risk compensation}}\right].$$

Depreciation condition

$$\mu'(u) < \sigma'(u)G(z).$$

- For $K > 0$, adverse expected-value news reinforces depreciation.
- Better expected fundamentals can offset the risk adjustment.
- A real-world TPU innovation may change both moments, interest rates, exposures, and risk-bearing capacity.

Price evidence alone cannot identify the risk-compensation mechanism.

A local variance representation and risk-bearing capacity

For small aggregate risk exposure, $|z| = |KR\sigma/T| \ll 1$, use $\tanh(z) \simeq z$:

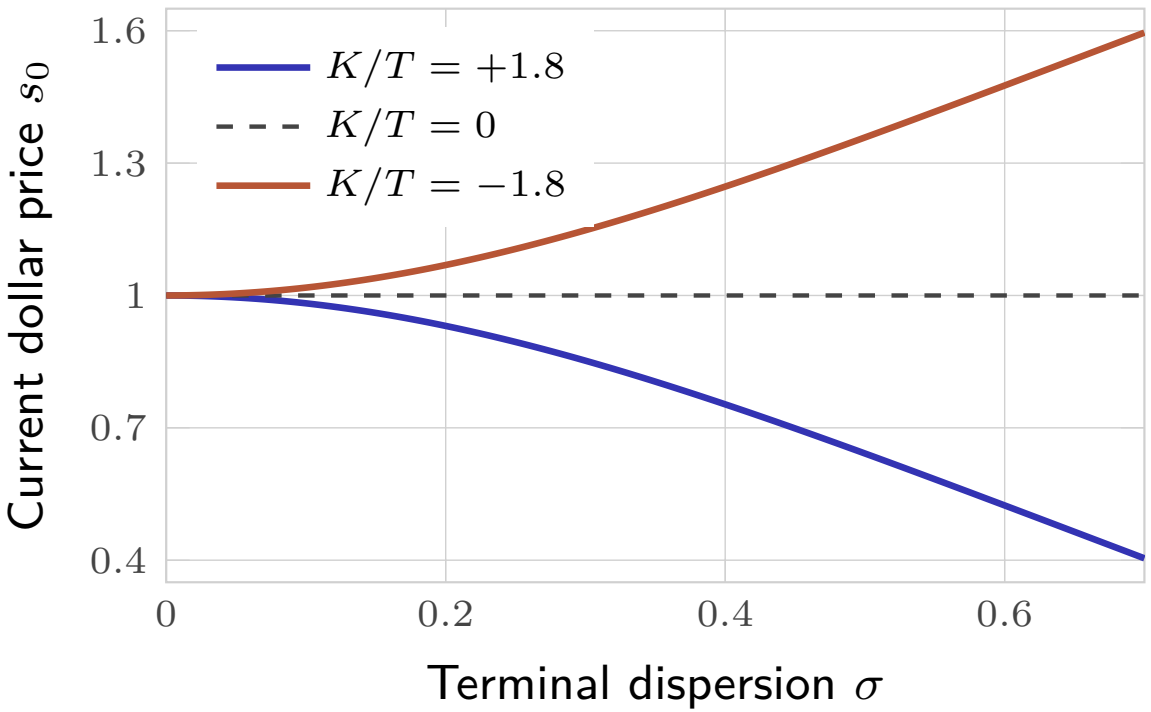
$$\rho \simeq \frac{K}{T} R^2 \sigma^2, \qquad s_0 \simeq \frac{R}{R^*}\mu - \frac{KR^2}{TR^*}\operatorname{Var}_0(\widetilde{s}_1).$$

The leading risk adjustment scales with exposure, payoff variance, and inverse risk tolerance.

$$\frac{\partial s_0}{\partial K} = -\frac{R^2\sigma^2}{R^*T}\operatorname{sech}^2(z) < 0, \qquad \frac{\partial s_0}{\partial T} = \frac{KR^2\sigma^2}{R^*T^2}\operatorname{sech}^2(z).$$

- More risk tolerance raises the peso price when $K > 0$.
- These are price-level effects. The exact model does not imply that more exposure always magnifies the marginal uncertainty response.

An illustrative equilibrium price response



Only net exposure changes across curves.

- $K > 0$: a risk discount.
- $K = 0$: no adjustment.
- $K < 0$: a hedge premium.

Each point is a separate equilibrium at a different dispersion.

Uncalibrated numerical illustration. Normalize $\mu = R = R^* = 1$ and set $K/T \in \{-1.8, 0, 1.8\}$. Curves use the exact equilibrium formula and do not describe a time path.

Price adjustment, observable implications, and limits

$$n_i^{\text{eq}} = \frac{K}{\gamma_i T} - h_i \quad \Longrightarrow \quad \frac{\partial n_i^{\text{eq}}}{\partial u} = 0 \quad \text{when } K, T, h_i \text{ are fixed.}$$

Existing portfolios can remain optimal after repricing. Financial pricing need not produce a new net trade.

Conditional implication	**Empirical qualification**
Currency risk rises	Daily realized risk is a proxy for terminal conditional risk.
Peso weakens for $K > 0$	Measure bilateral CLP/USD and assess expected-value news.
Expected dollar return rises	Requires maturity-matched rates and suitable expectations or returns.

Scope: no prediction for national current- or financial-account balances, or for multi-month IRF persistence.

Financial transactions and valuation changes are distinct in balance-of-payments accounting (IMF, BPM6, Chapter 9).

Preliminary empirical strategy and next steps

Core data. Chile, 2003M1–2024M9. Monthly TPU, month-end CLP/USD, daily returns, and BCCh surveys. W_{t-1} includes three monthly lags and lagged global/domestic controls.

$$
\begin{aligned}
U_t &= \ln(1 + TPU_t) = a + \Pi' W_{t-1} + \xi_t, \\
\eta_t &= \widehat{\xi}_t / \operatorname{sd}(\widehat{\xi}_t), \\
100[\ln E_{t+h} - \ln E_{t-1}] &= \alpha_h + \beta_h \eta_t + \Gamma_h' W_{t-1} + \epsilon_{t,h}, \\
q_{t+h} &= \delta_h + \theta_h \eta_t + \Lambda_h' W_{t-1} + v_{t,h}.
\end{aligned}
$$

$q_t = \ln[(252/n_t)\sum_{d \in t} r_d^2]$, $r_d = \Delta \ln E_d$; n_t counts daily returns. Main $h = 0, 1, 3$ months. Positive β_h indicates depreciation; positive θ_h higher risk.

Interpretation and inference

- η_t is a statistical innovation. Common global news can confound causality.
- 95% HAC intervals. A time-series bootstrap re-estimates the innovation.

Next steps

1. Audit dates, survey cutoffs, and forecast horizons.
2. Estimate price/risk LPs and survey expectation responses.
3. Extend to matched-maturity returns and event windows if feasible.

LP: Jordà (2005). Causal interpretation needs additional assumptions.